

Investigation of the structure of atomically dispersed NiN_x sites in Ni, N-doped carbon electrocatalysts by ⁶¹Ni Mössbauer Spectroscopy and Simulations

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Abstract

Ni and nitrogen-doped carbons are selective catalysts for CO₂ reduction to CO (CO₂R) but the hypothesized NiN_x active sites are challenging to probe with traditional characterization methods. Here, we synthesize ⁶¹Ni-enriched model catalysts, termed ⁶¹NiPACN, in order to apply ⁶¹Ni Mössbauer spectroscopy using synchrotron-radiation (⁶¹Ni-SR-MS) to characterize the structure of these atomically dispersed NiN_x sites. First, we demonstrate that the CO₂R results and standard characterization techniques (SEM, PXRD, XPS, XANES, EXAFS) point to the existence of dispersed Ni active sites. Then, ⁶¹Ni-SR-MS reveal significant internal magnetic fields of ~5.4 T which is characteristic of paramagnetic, high spin Ni²⁺, in the ⁶¹NiPACN samples. Finally, theoretical calculations for a variety of Ni-N_x moieties confirm that high spin Ni²⁺ is stable in non-planar, tetrahedrally-distorted geometries which result in calculated isotropic hyperfine coupling that are consistent with ⁶¹Ni-SR-MS measurements.

Introduction

Metal and nitrogen-doped carbons (M-N-Cs) are a widely studied class of materials due to the atomically dispersed, nitrogen-coordinated MN_x sites that are hypothesized to be catalytic active sites for a variety of chemical reactions.¹ In particular, Ni-N-C catalysts show high selectivity for CO under electrochemical CO₂ reduction conditions^{2,3} and Fe-N-C catalysts have been investigated for their activity (ORR) in both acidic and alkaline environments.⁴

There has been considerable research interest in identifying the exact structure of the catalytic active site in M-N-Cs to facilitate the design of optimized, next-generation M-N-C catalysts. However, the presumed active MN_x sites are challenging to characterize due to their lack of long-range order which prevents the use of diffraction-based techniques and because of the wide

distribution of metal environments in M-N-C including metal, oxide, carbide, and nitride species.⁵ To address this challenge, past work on the Fe-N-C system has demonstrated that combining a wide variety of techniques to obtain a self-consistent picture of the catalyst surface can yield new information on the catalytic active site. For example, researchers have leveraged insights from X-ray photoelectron spectroscopy (XPS)⁶, X-ray absorption spectroscopy (XAS) including the near-edge (XANES) and extended fine structure (EXAFS)^{7,8}, high angle annular dark field imaging with a scanning tunneling electron microscope (HAADF-STEM)⁹, electron energy loss spectroscopy (EELS)⁹, time-of-flight secondary ion mass spectrometry (ToF-SIMS)¹⁰ and ⁵⁷Fe Mossbauer spectroscopy.¹¹ This combination of techniques has conclusively pointed towards FeN₄ ORR active sites though contributions from other FeN_x sites is also possible.

However, even within the class of ‘FeN₄ sites’ there can be significant structural diversity due to changes in the secondary coordination sphere, distortions in the molecular geometry of the FeN₄ complex, or differences in spin-state.¹² One of the few techniques proven to quantitatively distinguish between different types of FeN₄ sites has been ⁵⁷Fe Mössbauer spectroscopy.^{11,13} Mössbauer spectroscopy, based on the absorption of light by compatible ‘Mössbauer-active’ atomic nuclei, is sensitive to subtle changes in electronic structure. Due to the particularly attractive qualities of probing the ⁵⁷Fe nucleus, ⁵⁷Fe Mössbauer spectroscopy has been applied with great success to Fe-N-C materials and helped assign the low spin FeN₄ and high spin N-FeN₂₊₂ sites as ORR active sites.¹¹

Though Fe-N-C has been the most widely studied M-N-C material, new applications for other M-N-Cs are gaining academic interest. One such system is Ni-N-C which has shown high selectivity and activity for electrocatalytic CO₂ reduction to CO (CO₂R) over the competing hydrogen evolution reaction (HER).^{3,14} Recent work has shown that analogous NiN_x sites are likely

CO₂R active sites, however, none of the characterization techniques employed have been able to ascertain the specific geometry and coordination of the active site.^{2,15–17} At the current state of knowledge, the existence of atomically dispersed, nitrogen-coordinated Ni sites has been clearly indicated but further investigation is needed to precisely define the site structure in these materials.

In this study, we employ ⁶¹Ni Mössbauer spectroscopy of Ni-N-C catalysts to provide additional depth in our understanding of the structure of NiN_x CO₂R active sites. To perform this study, we devised a synthesis strategy to make isotopically-enriched ⁶¹Ni-N-C catalysts which could be tested using ⁶¹Ni synchrotron-radiation Mössbauer spectroscopy (SR-MS) which has been recently developed at the SPring-8 synchrotron (Hyogo, Japan).¹⁸ To supplement these ⁶¹Ni Mössbauer spectra, we also performed a Density Functional Theory (DFT) simulation to calculate hyperfine coupling parameters on Ni-N_x moieties. In addition, extensive characterization including CO₂R testing, XAS, XPS, and PXRD was used to corroborate the conclusions about the structure of NiN_x sites. This holistic approach, employing the emerging ⁶¹Ni SR-MS technique alongside complementary structural characterization methods, provides a self-consistent picture of the structure of Ni-N-C catalysts which can inform future CO₂R catalyst design efforts.

Methods

Catalyst Synthesis: ⁶¹NiPACN samples were synthesized by a modified version of the procedure reported previously for NiPACN with all materials purchased from Sigma Aldrich unless otherwise noted. Acrylonitrile (2 mL) and azobisisobutyronitrile (AIBN, 2 mg) in acetone (2 mL) was reacted overnight at 70°C in an oven to form white polyacrylonitrile powder. Then, an oxidative treatment (230°C, 0.1°C/min, 2 hr hold, air) and a pyrolysis treatment (900°C, 5°C/min, 2 hr hold, N₂) were used to synthesize black, carbonized PACN powder (metal-free). Then, ⁶¹Ni metal powder (16 mg, Isoflex, 99.42%) was dissolved in 1M nitric acid (10 mL). An appropriate

amount of the ^{61}Ni nitric acid solution was added to a dispersion of PACN in 1M nitric acid to achieve a desired nominal Ni weight loading. The dispersion was sonicated for 1 hr and then heated while stirring on a hot plate at 100°C until dried. The black powder was collected and re-pyrolyzed (900°C , $5^\circ\text{C}/\text{min}$, 2 hr hold, N_2) to form the $^{61}\text{NiPACN}$ catalysts. Two batches of the $^{61}\text{NiPACN}$ samples were made with the first batch tested with ^{61}Ni -SR-MS, Ni K-edge XAS, PXRD, and CO_2R catalysis and the second batch for XPS.

CO_2R Testing: Electrochemical testing for CO_2 reduction was performed in a polycarbonate 2-compartment cell reported previously (1 cm^2 working area, 1.8 mL catholyte and anolyte volumes separated by a membrane (Selemion, AMV;AGC)). A glassy carbon counter electrode (Alfa Aesar, 2 mm thickness) and leak-free Ag/AgCl reference electrode (ET072-1, eDAQ) were used. The working electrode was prepared by spraying (Speedaire 4RR09J, 30 psi, N_2) a $^{61}\text{NiPACN}$ ink (12.5 mg catalyst in 1 mL EtOH with 25 μL of 5 wt% Nafion solution, sonicated for 1 hr) onto glassy carbon at a desired catalyst loading. All experiments reported here were performed in 0.1M KHCO_3 with 10 sccm CO_2 sparged through both chambers and in-line gas chromatography (SRI Instruments; 8610C) for product quantification.

^{61}Ni Synchrotron-Radiation Mössbauer spectroscopy (^{61}Ni -SR-MS): ^{61}Ni -SR-MS experiments were performed largely following the methods outlined in a prior report.¹⁸ Samples were loaded into quartz tubes (Technical Glass Products, 2 mm ID, 3 mm ID). Measurements were taken at BL09XU at SPring-8 in the “203 bunch” A-mode with 23.6 ns electron bunch spacing and the beam monochromated to 67.4 keV using a Si (333) high heat load monochromator and then a Si (111) low-energy filtering monochromator. Transmitted photons were quantified using a $^{61}\text{Ni}_{0.86}\text{V}_{0.14}$ analyzer foil mounted on a velocity transducer for selecting energies by Doppler shift. The velocity axis was calibrated by simultaneous laser interferometry. Samples and the detector

foil were cooled to ~ 5 K with liquid He during the experiments. Least-squares fits were performed in Matlab using 'lsqcurvefit' for a magnetic Hamiltonian and a magnetic plus perturbing quadrupolar Hamiltonian (resulting in the component of electric field gradient in the direction of the magnetic field).¹⁹ All fits were performed after subtracting a polynomial background (to account for the time-windowing effect inherent in SR-MS^{20,21}) and kept a constant FWHM for all Lorentzian-broadened peaks of 0.8 mm/s with fitting parameters of 1) intensity, 2) B and, in some cases, 3) V_{zz} . Standard error of fit parameters were calculated using the "nlparci" function in Matlab.

Physical and Chemical Characterization: Powder X-ray diffraction (PXRD, Bruker D8 Venture) was measured on powder samples in Kapton capillaries with Cu K α radiation. X-ray photoelectron spectroscopy (XPS, PHI Versaprobe 3) was performed with Al K α radiation with electron and ion neutralization and pass energies of 55 eV for C and N and 225 eV for survey scans and additional Ni scans. Ni K-edge X-ray absorption spectroscopy (XAS, BL 11-2, SSRL, SLAC National Accelerator Laboratory) was performed on powders adhered to Kapton tape with fluorescence detection using a 100-pixel solid state Ge detector. The incident beam was monochromated by a Si (220) liquid-nitrogen cooled double-crystal monochromator with the energy axis calibrated by the first inflection point of a downstream nickel foil set to 8333 eV. Data was collected between 8100 eV and 9200 eV. For EXAFS data, Fourier-transforms were performed with a k-range of [2 $\text{\AA}^{-1}$ 8 $\text{\AA}^{-1}$] with k^2 weighting. No time-dependent spectral changes were observed with a repeated data collection.

Theoretical Calculations Details: The molecular calculations of the hyperfine coupling parameters have been extensively tested in a variety of transition-metal complexes.^{22,23} Electron paramagnetic resonance (EPR) calculations were conducted to find isotropic hyperfine coupling constants for Ni

using NiN_4C_8 and $\text{NiN}_4\text{C}_{10}$ clusters in ORCA program package (version 5.0.0).²⁴ We used B3LYP²⁵ density functional with SV(P) for all elements and TZV(P) basis set for Ni for geometry optimizations.²⁶ Furthermore, we used def2/J auxiliary basis sets²⁷ for all elements together with CP(PPP) basis set²⁸ for Ni used during single point calculations. Finally, calculations of the isotropic hyperfine A tensor (A_{iso}) and electron densities at the Ni nuclei were performed using the EPR-II basis set²⁹ and the zero-order regular approximation (ZORA).³⁰ The spin densities at Ni were calculated using Multiwfn wavefunctional analyzer.³¹

For energy comparison, we used spin-polarized DFT calculations for geometry optimization using the Perdew–Burke–Ernzerhof (PBE)³² exchange-correlation functional of generalized gradient approximation³³ as implemented in the Vienna ab initio simulation package (VASP).³⁴ The energy cutoff of the plane wave basis set was 500 eV for all VASP calculations. We applied DFT+ U method using $U_{\text{eff}} = 6.45$ eV for Ni 3d orbital to correct artificial electron self-interaction.^{35,36} For geometry optimization and formation/binding energy calculation, we used 4×4 graphene super cell containing NiN_4 (except NiN_3O_1 , NiN_2O_2) moiety in basal plane. We utilized a 20 Å vacuum gap to remove periodic interaction in perpendicular direction. The k-points mesh were chosen to $5 \times 5 \times 1$ at the Brillouin zone using the Monkhorst–Pack scheme.³⁷ The geometry optimization criteria were set to 10^{-6} eV for energy convergence and 0.01 eV Å⁻¹ for forces on each atom respectively. The high spin state was achieved by setting initial magnetic moments on the Ni atom, followed by the geometry optimization. For binding energy calculations, we used H_2 , O_2 , and NO and CO molecules and graphene system for reference energies.

Results and Discussion

Synthesis of $^{61}\text{NiPACN}$:

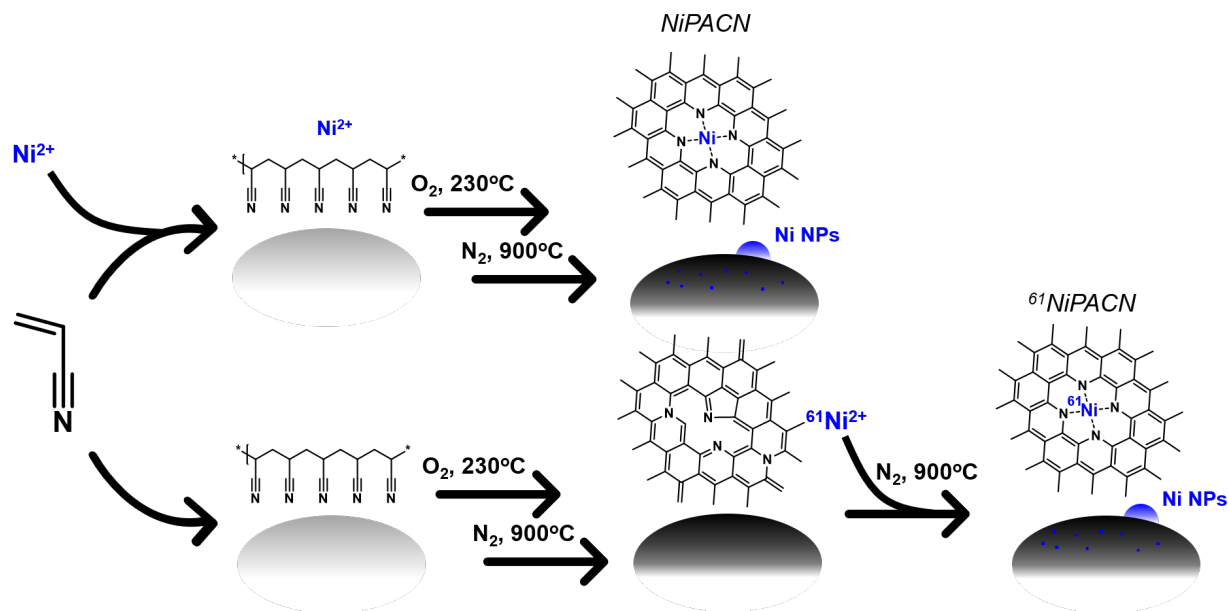


Figure 1. Synthesis schematic showing previously published synthesis of NiPACN (upper half) and modified procedure used in this publication for $^{61}\text{NiPACN}$ (lower half)

In order to use ^{61}Ni SR-MS to probe Ni-N-C active sites, a suitable ^{61}Ni enriched catalyst sample needed to be synthesized. To achieve this, we followed a synthesis protocol used to make NiPACN, a polyacrylonitrile-derived Ni-N-C material, that was reported by our groups recently (**Figure 1**).¹⁵ However, the ^{61}Ni precursor material could only be purchased in metallic form which necessitated a modification to the NiPACN synthesis which used acetone-soluble nickel nitrate as the Ni source.¹⁵ For $^{61}\text{NiPACN}$, we first synthesized metal-free N-doped carbon by the traditional polymerization, oxidation, and pyrolysis steps and then deposited Ni onto the powder by dispersion in an acidic solution that could dissolve the metallic ^{61}Ni precursor. Then, after drying and a second pyrolysis step, the ^{61}Ni -enriched $^{61}\text{NiPACN}$ samples could be used for further testing and characterization (**Figure 3**). In this study, three $^{61}\text{NiPACN}$ samples were synthesized and denoted 0.1 wt%, 0.5 wt%, and 1 wt% (based on Ni addition during synthesis) along with a control 0 wt%

sample that was subjected to the acid treatment and 2nd pyrolysis without addition of ⁶¹Ni. We note that these denominations are significant overestimates of the true Ni loading measured by inductively coupled plasma atomic emission spectroscopy (ICP-AES, **SI-Table 1**) but the naming convention based on the amount of Ni added during synthesis was kept for brevity. In addition, two batches of each of the ⁶¹NiPACN catalysts were synthesized; the first batch was used for ⁶¹Ni-SR-MS, Ni K-edge XAS, PXRD, and CO₂R catalysis and the second for XPS. The ICP-AES showed qualitatively similar trends in Ni loading between the two batches but not quantitative matches, thus all conclusions herein are based on the qualitative Ni chemical states identified by physical characterization.

These ⁶¹NiPACN samples were representative of the NiPACN materials that have already been investigated by our group by a variety of methods including XAS, PXRD, XPS, HAADF-STEM, EELS, and ToF-SIMS.^{15,16} However, as the ⁶¹NiPACN synthesis method diverges slightly from the NiPACN method we first aimed to demonstrate that there is evidence for NiN_x active sites before proceeding to ⁶¹Ni SR-MS.

CO₂ Reduction:

Electrochemical CO₂ reduction experiments were conducted to determine the catalytic activity of ⁶¹NiPACN for CO₂R to CO and HER (**Figure 2**). Because NiN_x sites are expected to be selective for CO₂R to CO¹⁵ and metallic Ni is known to be nearly completely inactive for CO₂R³⁸, the relative activities for CO₂R and HER are an indirect measurement of the distribution of Ni-containing sites in ⁶¹NiPACN. The CO₂R experiments were performed in a 2-compartment electrochemical cell with a catalyst loading of 0.2 mg/cm² on the glassy carbon working electrode.³⁹

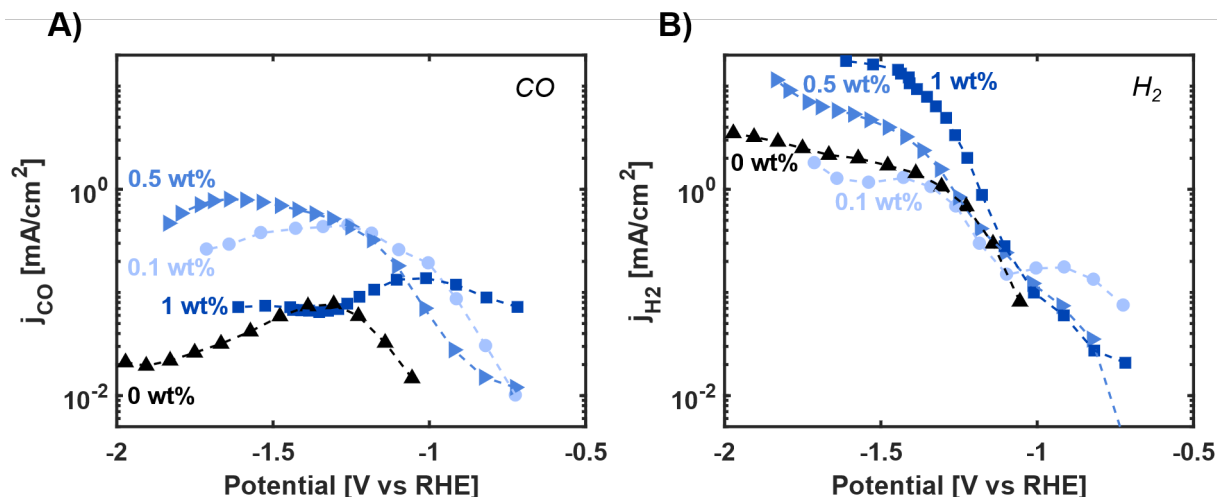


Figure 2. (A) Partial current density (log axis) to CO versus electrode potential for $^{61}\text{NiPACN}$ samples with nominal Ni weight loadings from synthesis of 0, 0.1, 0.5, and 1 wt% (labeled directly on plot) and (B) corresponding partial current densities to H_2 . All experiments in 2-compartment cell with 0.1 M KHCO_3 electrolyte and 10 sccm CO_2 input flowrate.

The CO_2R results show that increasing the ^{61}Ni content in the catalyst from the 0 wt% to 0.1 wt% samples (nominal Ni weights from synthesis) greatly increased the activity for CO_2R without significant effect on HER rates. Thus, at this dilute Ni concentration, the dominant Ni species are dispersed Ni sites that are likely CO_2R active sites. The j_{CO} versus potential shows a characteristic plateau of $\sim 0.3 \text{ mA}/\text{cm}^2$ at potentials near -1.1 V vs RHE which corresponds to the onset of a mass transport limited current density (**Figure 2A**). In this case, the total number of available CO_2R active sites must be significantly lower than necessary to satisfy the flat-plate limiting current of $\sim 6 \text{ mA}/\text{cm}^2$ that was achieved in this cell using silver foils. This suggests that the limiting j_{CO} current is correlated to the number of CO_2R active sites. The activity of the $^{61}\text{NiPACN}$ -0.5 wt% sample supports this hypothesis because the limiting current rose to $\sim 0.8 \text{ mA}/\text{cm}^2$ is observed which corresponds to an increase in dispersed Ni active site content on the electrode. The HER performance of the 0.5 wt% sample shows higher current than the 0.1 wt% and 0 wt% samples which suggests that, despite the increase in dispersed Ni active site content, there is also the possibility of Ni aggregate or other HER-active Ni species formation in this sample

(**Figure 2B**). The activity of the highest loaded sample, $^{61}\text{NiPACN-1 wt\%}$, shows different electrochemical CO_2R behavior relative to the two lower loading samples. This 1 wt% sample has negligible CO_2R activity, below 0.1 mA/cm^2 , and has further increased HER rates relative to the 0.5 wt% sample. Thus, at this higher loading it is likely that the sample is dominated by Ni aggregates that catalyze HER and not CO_2R and very few catalytically relevant dispersed Ni sites are formed. In summary, the CO_2R testing suggests that the $^{61}\text{NiPACN}$ samples with nominal Ni loadings of 0.1 wt% and 0.5 wt% contain the dispersed NiN_x active sites while the 1 wt% sample contains mostly HER-active Ni aggregates.

We note that this relationship between CO_2R activity and Ni loading is different between $^{61}\text{NiPACN}$ and NiPACN . As reported in our earlier publication, NiPACN shows an increase in j_{CO} (at a given electrode potential) with increasing Ni loading up to $\sim 2 \text{ wt\% Ni}$ after which the j_{CO} plateaus. In contrast, the $^{61}\text{NiPACN}$ samples do not seem to have a plateau in j_{CO} and instead a decline in CO_2R activity is observed at the highest loading. We hypothesize that this difference arises from the different synthetic methods used for the catalysts. In particular, the $^{61}\text{NiPACN}$ samples were made by adsorbing Ni^{2+} onto only the surface of a N-doped carbon substrate in contrast to the homogeneous dispersion of Ni^{2+} achieved in NiPACN by adding the Ni precursor during the polymerization. During the pyrolysis step, the confinement of Ni^{2+} to the surface for $^{61}\text{NiPACN}$ might lead to a higher propensity for interaction between Ni atoms and subsequent aggregation into Ni nanoparticles.

Evidence for dispersed Ni sites:

Before analyzing ^{61}Ni Mössbauer spectra of the $^{61}\text{NiPACN}$ samples, we first used standard physiochemical characterization techniques to probe the structure of the dispersed Ni moieties that were identified as CO_2R active sites from catalytic characterization.

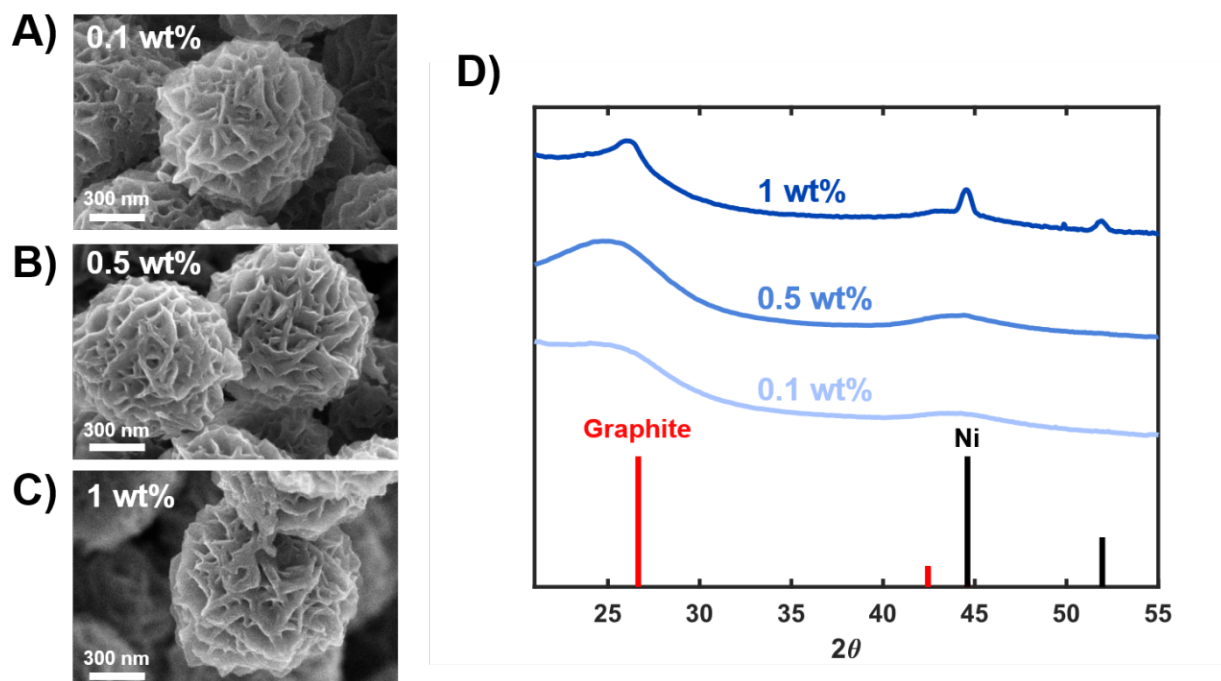


Figure 3. SEM images of (A) $^{61}\text{NiPACN}$ -0.1 wt%, (B) $^{61}\text{NiPACN}$ -0.5 wt%, (C) $^{61}\text{NiPACN}$ -1 wt%. (D) PXRD patterns of reference graphite (red) and metallic fcc Ni (black) and three $^{61}\text{NiPACN}$ samples.

Scanning electron microscopy (SEM) shows the ‘flower-like’ morphology for all of the $^{61}\text{NiPACN}$ samples (**Figure 3A-3B**). This morphology is characteristic of this synthesis that uses polyacrylonitrile particles as the carbon-precursor⁴⁰ and the samples show no obvious differences as a function of Ni loading.

Powder X-ray diffraction (PXRD) was used to confirm the existence of Ni aggregates that form in the $^{61}\text{NiPACN}$ samples (**Figure 3D**). PXRD patterns show broad, graphitic features at 25° and 44° for all three $^{61}\text{NiPACN}$ samples consistent with carbonization during pyrolysis treatments. No other features are present for the 0.1 wt% and 0.5 wt% samples but the 1 wt% samples have metallic Ni peaks at 44.5° and 52°. In addition, the graphitic features on all three samples became monotonically sharper with increased Ni loading in agreement with prior findings. As metallic Ni is a catalyst for graphitization, this suggests that the metallic Ni content of the samples also increases with Ni loading though the amount of metallic Ni in the 0.5 wt% sample is below the

detection limit of PXRD. These PXRD results are consistent with the monotonic increase of HER with loading and suggest that dispersed sites are dominant in the 0.1 wt% and 0.5 wt% samples while Ni aggregates dominate the 1 wt% sample.

X-ray photoelectron spectroscopy (XPS) provides additional information on the chemical states of the N and Ni dopants in $^{61}\text{NiPACN}$ (**SI-Figure 1**). The N1s region of all samples can be deconvoluted into characteristic peaks at 398.2 eV, 401.1 eV, and 403.6 eV, corresponding to pyridinic, graphitic, and oxidized N respectively.⁴¹ The fraction of N into these three categories of N defects does not substantially change between the samples (**SI-Table 2**) which is consistent with very small amounts of Ni incorporation that doesn't substantially affect the signal from the relatively large XPS probe depth compared to the surface defects. The Ni 2p region shows only very weak signal for the 0.1 wt% sample due to the Ni content being below the typical detectability limit of XPS. The 0.5 wt% sample shows a clear peak at 854.6 eV, corresponding to Ni^{2+} , which is consistent with the oxidation state of NiN_x states reported in prior publications.⁴² The 1 wt% sample has a small feature at the same energy, ~ 855 eV, corresponding to Ni^{2+} , which conflicts with the existence of Ni nanoparticles and higher Ni content implied by the preceding analysis. We hypothesize that the carbon shells that form around Ni particles under pyrolysis conditions would both attenuate XPS signal and limit the probe depth to the near-surface region of the particles where a native Ni oxide layer would form and give Ni^{2+} signal.

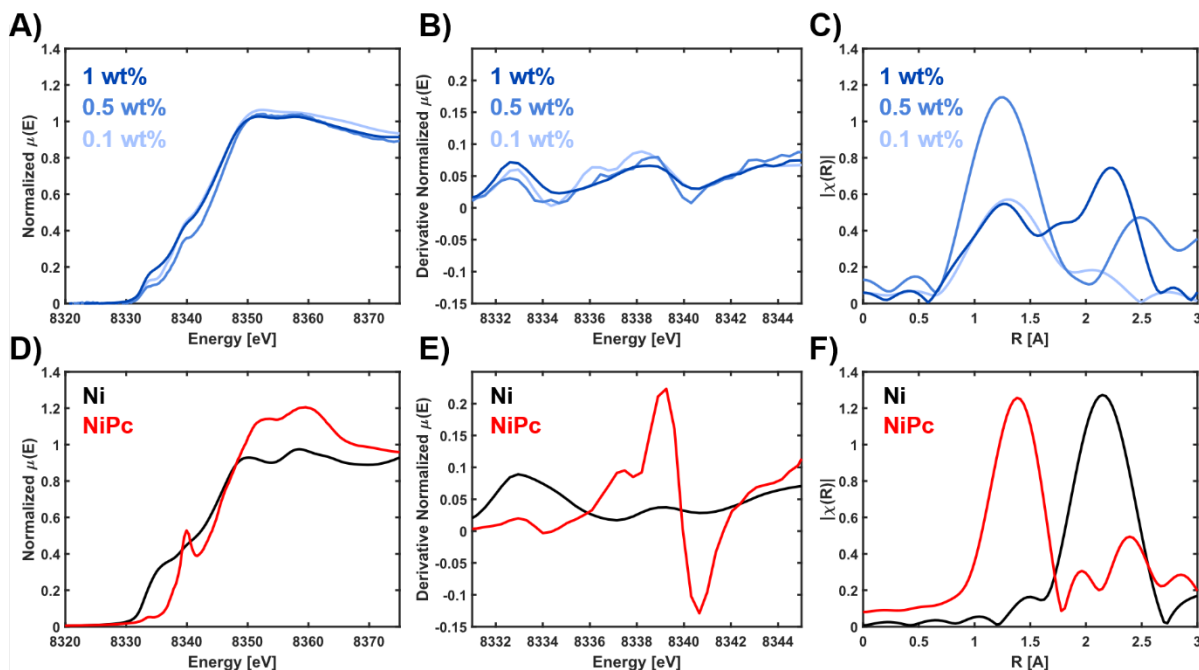


Figure 4. Ni K-edge XAS of $^{61}\text{NiPACN}$ samples with (A) normalized XANES, (B) First Derivative XANES, and (C) EXAFS. Corresponding XAS of reference spectra for metallic fcc Ni (black) and Ni Phthalocyanine (NiPc, red) with (D) normalized XANES, (E) First Derivative XANES, and (F) EXAFS (not phase corrected).

Finally, X-ray absorption spectroscopy (XAS) at the Ni K-edge was performed to assess the chemical state and bonding environment of the Ni atoms in the $^{61}\text{NiPACN}$ samples (**Figure 4**). The near-edge structure of the XAS spectra (XANES) is useful for chemical fingerprinting of oxidation state and geometry. XANES of the $^{61}\text{NiPACN}$ samples show features at 8334 eV and 8340 eV and a small shift of the edge to lower energies for the 1 wt% sample, consistent with a more reduced Ni environment (**Figure 4A**). The spectrum of a reference molecular square planar complex, Ni Phthalocyanine (NiPc), shows the existence of a $1s \rightarrow 4p$ feature at 8340 eV which is characteristic of square planar geometries and a dipole forbidden $1s \rightarrow 3d$ peak at 8334 eV which has higher intensity in reduced symmetry environments (**Figure 4D**).⁴³ The derivative XANES spectra demonstrate that the $1s \rightarrow 4p$ feature (8340 eV) exists in the $^{61}\text{NiPACN}$ samples which supports the existence of a square planar-like geometry of Ni atoms in agreement with prior reports

(**Figure 4B, 4E**).^{15,44} The extended XAS fine structure (EXAFS), originating from the scattering of photoelectrons off neighboring atoms, gives direct information of the bonding environment in the samples (**Figure 4C, 4F**, k-space plot in **SI-Figure 2**). The main prominent feature for the 0.1 wt% sample is at 1.29 Å which we match to the Ni-N scattering peak for NiPc at 1.38 Å. The 0.5 wt% also shows this Ni-N peak and a second peak at 2.48 Å that likely corresponds to carbon atoms in the secondary coordination shell. Consistent with the prior observations, the 1 wt% sample shows a prominent peak at 2.22 Å which roughly matches the primary peak at 2.15 Å for metallic Ni. The 1 wt% sample also has the first coordination peak at 1.26 Å which could represent either coexisting dispersed NiN_x sites or oxidized Ni that exists along with metallic Ni.

Using an array of standard characterization techniques including SEM, PXRD, XPS, and XAS we show that the existence of dispersed NiN_x sites exist as the predominant form of Ni in the 0.1 wt% and 0.5 wt% Ni loaded samples and that metallic Ni becomes more prevalent in the 1 wt% sample in direct agreement with the CO₂R catalytic results.

⁶¹Ni Mössbauer Spectroscopy:

The existence of NiN_x CO₂R active sites is implied by the preceding analyses; we next used ⁶¹Ni synchrotron radiation-based Mössbauer spectroscopy (⁶¹Ni-SR-MS) in the energy-domain to further characterize the structure of these dispersed NiN_x sites. These measurements used the ⁶¹Ni-SR-MS instrumentation and methodology that was recently reported¹⁸; the three ⁶¹NiPACN samples were measured along with a metallic Ni powder for verification of the experimental methods. The ⁶¹Ni-SR-MS spectra of the metallic Ni powder showed a broad and symmetric peak characteristic of magnetic Zeeman splitting and could be fit to an internal magnetic field of 7.3 T (**SI-Figure 3**). This value agrees well with reference values of ~7.5 T^{45,46} and validates the application of ⁶¹Ni-SR-MS towards the analysis of the ⁶¹NiPACN powders.

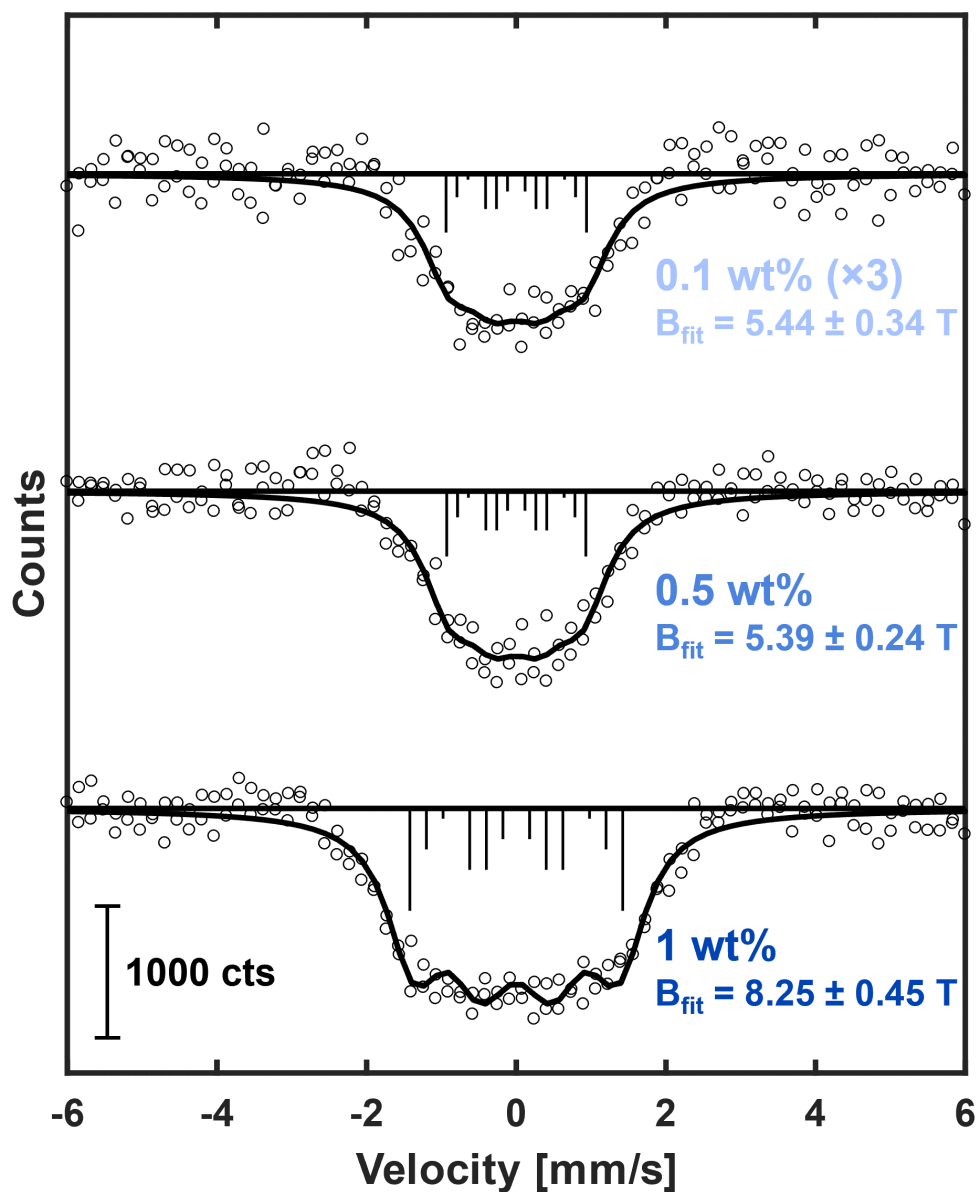


Figure 5. ^{61}Ni Mossbauer spectra of $^{61}\text{NiPACN}$ -0.1 wt% (upper), $^{61}\text{NiPACN}$ -0.5 wt% (middle), and $^{61}\text{NiPACN}$ -1 wt% (lower) with raw data shown as open black circles, fit to data using magnetic Zeeman splitting Hamiltonian in solid black lines, and individual transition locations and intensities as vertical black lines. Spectra taken at $T = 5$ K.

^{61}Ni -SR-MS spectra of all three $^{61}\text{NiPACN}$ powders showed a single symmetric, broad peak that is indicative of a magnetic hyperfine interaction that splits the spectra into 12 transitions (Figure 5).¹⁸ Fitting each spectrum to a single site, magnetic Hamiltonian provides an estimate of

the internal hyperfine magnetic field, B_{fit} , for the samples. Slight anisotropy can be observed on the $^{61}\text{NiPACN-0.1 wt\%}$ sample which could indicate a relevant electric field gradient.¹⁹ Incorporation of a perturbative quadrupolar term in the Hamiltonian along the axis of the internal field did not yield significantly improved fits for any of the samples and thus only the single site magnetic Hamiltonian was used for subsequent analysis (**SI-Figure 4, SI-Table 3**).

These fits revealed internal magnetic fields of $\sim 5.4 \pm 0.4$ T for the 0.1 wt% and 0.5 wt% samples and a larger field of 8.25 ± 0.45 T for the 1 wt% sample (**Figure 5**). These results qualitatively match the catalytic and physiochemical characterization where the two dilute samples are expected to have similar, dispersed sites and the 1 wt% sample should have distinct Ni aggregate-like behavior. As metallic Ni is expected to have internal fields of $\sim 7\text{-}8$ T and NiO of ~ 10 T,⁴⁷ the spectra and fitted field of 8.2 T for the 1 wt% sample is consistent with the existence of metallic Ni particles with oxide shells. The smaller value of 5.4 T for the 0.1 wt% and 0.5 wt% is indicative of a distinct chemical environment and likely arises from the bonding configuration in the NiN_x ligand field.

Due to the relative rarity of the ^{61}Ni -MS technique, reports on molecular Ni complexes that are dominated by a magnetic interaction are difficult to find which makes it challenging to contextualize the 5.4 T field. However, the existence of this hyperfine magnetic field is already indicative that the NiN_x sites in $^{61}\text{NiPACN}$ have nonzero spin density at the nucleus that couples with the nuclear spin through the.¹⁹ In other words, ^{61}Ni -SR-MS suggests that the NiN_x sites are paramagnetic or, in the case of Ni^{2+} , are high spin ($S = 1$) which leads to spin polarization of core orbitals⁴⁸ and results in the Fermi contact interaction. This is in opposition to the commonly accepted square planar geometry of the NiN_x sites which are unlikely to have a high spin ground

state due to the high-lying $d_{x^2-y^2}$ orbital.⁴⁹ Thus, the existence and origin of paramagnetism in the NiN_x sites in ⁶¹NiPACN need to be understood for a more accurate description of the active site.

Calculations of the hyperfine coupling tensor

Next, we used density functional theory (DFT) calculations to better understand the origin of paramagnetism in the NiPACN materials. We used two distinct NiN₄C₈ and NiN₄C₁₀ model structures to represent the dispersed NiN_x CO₂R active sites.^{50,51} In the NiN₄C₈ model, the NiN₄ moiety acts as a bridge between two adjacent armchair edges of graphene allowing NiN₄ to be tetrahedrally distorted (D_{2d} symmetry), creating two intersecting planes under a dihedral angle between 180° and 90° (**Figure 6A and SI Figure 5A**). For the NiN₄C₁₀ model, we studied both non-periodic and periodic configurations, where the square-planar (D_{4h} symmetry) NiN₄ moiety is embedded on a flat graphite layer (**SI Figure 5B, C**).

In general, after a large investigation of periodic and non-periodic Ni-N-C models in VASP and ORCA code correspondingly, we have found that all the square planar structures strongly prefer (~ -1 eV) a low spin (LS) state ($\langle S \rangle = 0$, singlet) than any other spin state, and resist out-of plane deformations of the Ni-center, regardless of the DFT+U or hybrid-DFT approach used (see *Theoretical Calculations Details*). Only when we allow for tetrahedrally distorted NiN₄ structures (D_{2d} symmetry), Ni energetically prefers (~ -0.2 eV) to be in a high spin (HS) state ($\langle S \rangle = 1$), as shown in **Figure 6A**. In this figure, we also show that optimal structures consist of dihedral angle between planes 153° to 123°. The low spin state of Ni is energetically more favorable over high spin, only for a square-planar geometry (dihedral angle of 180°).

Finally, we calculated isotropic hyperfine coupling constants (A_{iso}) of the above systems to compare to the experimental Mössbauer spectra (see *Theoretical Calculations Details*). This

hyperfine coupling constant is a component of the hyperfine coupling tensor, A , that is used in the spin Hamiltonian formalism to fit EPR and Mössbauer spectra. Resolving the full tensor using the experimental Mössbauer spectra is challenging for the powdered NiPACN samples, instead the fitted internal magnetic field value of 5.4 ± 0.3 T can be converted to a hyperfine coupling constant. To do so, the internal field units are converted using the nuclear g factor of ^{61}Ni (-0.5), the nuclear magneton (7.62 MHz/T) and making the limiting assumption that $\langle S \rangle$, the expectation value of electronic spin, is 1 as defined by Equation $A_{exp} = -B_{fit} g_N \mu_N / \langle S \rangle$.^{19,52} This results in a hyperfine coupling constant $A_{exp} = 20.6 \pm 1.1$ MHz which gives an upper bound estimate for comparison to DFT calculations.

To start, we have benchmarked our DFT approach for $\text{NiH}(\text{CO})_3$ system which has experimental $A_{iso} = 9$ MHz⁵³.⁵³ Our calculated A_{iso} gives value of 8.81 MHz and is also similar to $A_{iso} = 5.2$ MHz reported previously.⁵⁴ Next, we found that NiN_4 square-planar geometry with constrained HS ($\langle S \rangle = 1$) to have a very high A_{iso} value ~ 44 MHz, while by definition, ($\langle S \rangle = 0$) system has a zero A_{iso} value. Using the NiN_4C_8 model, which is allowed to undergo tetrahedral distortion NiN_4 (**Figure 6B**), its A_{iso} value monotonically decreases with decreasing dihedral angle. For the optimal angles between 153° to 123° , we obtain A_{iso} values ranging from 37.56 to 26.15 MHz. Such values are approaching the experimental upper bound of $A_{exp} = 20.6 \pm 1.1$ MHz for $^{61}\text{NiPACN}$ -0.5 wt% samples. Further distortion from square-planar to almost fully tetrahedral geometry (92.5° dihedral angle) yields a low A_{iso} value of 9.19 MHz.

We note that another route towards achieving a structural distortion that induces high spin character is through adsorbates that induce a square-pyramidal geometry. This class of site structures were also computationally investigated for a variety of adsorbates which showed that

some adsorbates could also induce hyperfine coupling constants in the realm of experimental values (see **SI-Discussion**, **SI-Figure 6**, **SI-Table 4, 5**).

Finally, we have extracted total and spin electron densities at the Ni nucleus and found good linear relationships with A_{iso} values for different distorted NiN_4 systems (**SI-Figure 7**). All other ferromagnetic properties for different studied systems are also tabulated in **SI-Table 5, 6**. Based on our results for relative energies and A_{iso} values, we conclude that a tetrahedral distortion of a NiN_4 site is a viable candidate for the site structure in Ni-N-C materials.

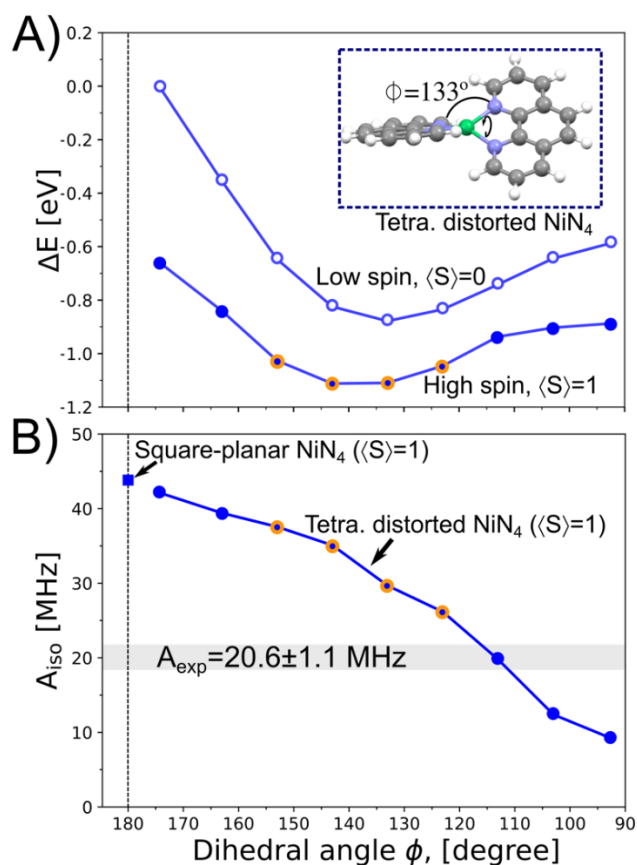


Figure 6. Electronic energy difference and isotropic coupling constant as a function of NiN_4 planes dihedral angle shown in the inset. A) The relative electronic energy difference for HS and LS configurations referenced to a 174° dihedral angle of NiN_4 distortion at low spin state. B) Calculated A_{iso} values as a function of NiN_4 planes dihedral angle (dash line at 180° angle indicates square-planar NiN_4 system).

Conclusion

Ni-N-C catalysts were synthesized with a ^{61}Ni source ($^{61}\text{NiPACN}$) to be probed by ^{61}Ni Mössbauer spectroscopy to deepen our understanding of dispersed, molecular-like NiN_x sites. CO_2 electroreduction and standard characterization techniques including SEM, XPS, PXRD, and XAS all confirm that these dispersed NiN_x sites are likely catalytic active sites and are the major form of Ni in 0.1 wt% and 0.5 wt% samples of $^{61}\text{NiPACN}$ while metallic Ni aggregates form in the 1 wt% sample. ^{61}Ni synchrotron radiation Mössbauer spectroscopy of the $^{61}\text{NiPACN}$ samples show a large magnetic interaction, $B_{\text{fit}} = 5.4 \text{ T}$, $A_{\text{exp}} = 20.6 \text{ MHz}$, at the Ni nuclei in the dilute 0.1 wt% and 0.5 wt% samples that suggests high spin Ni^{2+} character of the NiN_x active sites. This assignment of high spin Ni^{2+} is inconsistent with a square planar NiN_x geometry. Our quantum chemical calculations showed that Ni-N-C systems with tetrahedrally distorted NiN_4 geometries have a stable Ni high spin ($\langle S \rangle = 1$) state, while square-planar geometry always prefers for Ni to have a low spin. The calculated hyperfine coupling constants, A_{iso} , for such distorted NiN_4 geometries approach the experimentally observed value of $A_{\text{exp}} = 20.6 \pm 1.1 \text{ MHz}$. This study provides further chemical specificity on the structure of NiN_x sites in Ni-N-C CO_2R catalysts and motivates future work that utilizes advanced characterization techniques to achieve a more complete description of these complex active sites.

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Data Availability

A detailed list of calculated energies and structures is provided via the [catalysis-hub.org](https://www.catalysis-hub.org) repository (<https://www.catalysis-hub.org/publications/KoshyInvestigation2022>)⁵⁵ The fully optimized structures, reference structures, and their respective energies are also included as part of the [catalysis-hub.org](https://www.catalysis-hub.org) repository. All other study data are included in the article and/or supporting information.

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